

Proportional or Not?



Grade 7 • Mathematics
NC.7.RP.2

Identify and represent proportional relationships.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

A table has (x,y) pairs [(1, 6), (2, 9), (4, 15)]. Is the relationship proportional? Explain using ratios and write a rule.

Ratios y/x are 6, $9/2$, $15/4$. They differ. The starting value is not zero.



Build a second model together

A table has (x,y) pairs [(1, 8), (2, 16), (4, 32)]. Is the relationship proportional? Explain using ratios and write a rule.

Discuss: What should the first step find?



Your turn • 1

A table has (x,y) pairs [(1, 5), (2, 8), (4, 14)]. Is the relationship proportional? Explain using ratios and write a rule.

Represent → solve → explain → check



Your turn • 2

A table has (x,y) pairs [(1, 4), (2, 8), (4, 16)]. Is the relationship proportional? Explain using ratios and write a rule.

Represent → solve → explain → check



Your turn • 3

A table has (x,y) pairs [(1, 5), (2, 8), (4, 14)]. Is the relationship proportional? Explain using ratios and write a rule.

Represent → solve → explain → check



Your turn • 4

A table has (x,y) pairs [(1, 9), (2, 18), (4, 36)]. Is the relationship proportional? Explain using ratios and write a rule.

Represent → solve → explain → check



Exit question • Explain independently

A table has (x,y) pairs [(1, 13), (2, 22), (4, 40)]. Is the relationship proportional? Explain using ratios and write a rule.

Include a check. What would you teach someone else?



Teacher review and next steps

1. No; $y = 3x + 2$

2. Yes; $y = 4x$

3. No; $y = 3x + 2$

4. Yes; $y = 9x$

Exit: No; $y = 9x + 4$

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

